

CONTROL SYSTEMS LABORATORY

Experiment No:	6
Experiment Name:	STABILITY ANALYSIS
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Date:	31.05.2023

1. OBJECTIVES

The main purpose of software experiments is to introduce the computer-aided control system analysis and design. In this experiment, you will practice about stability analysis by using MATLAB.

2. EXPERIMENTAL RESULTS

A-

1,2-

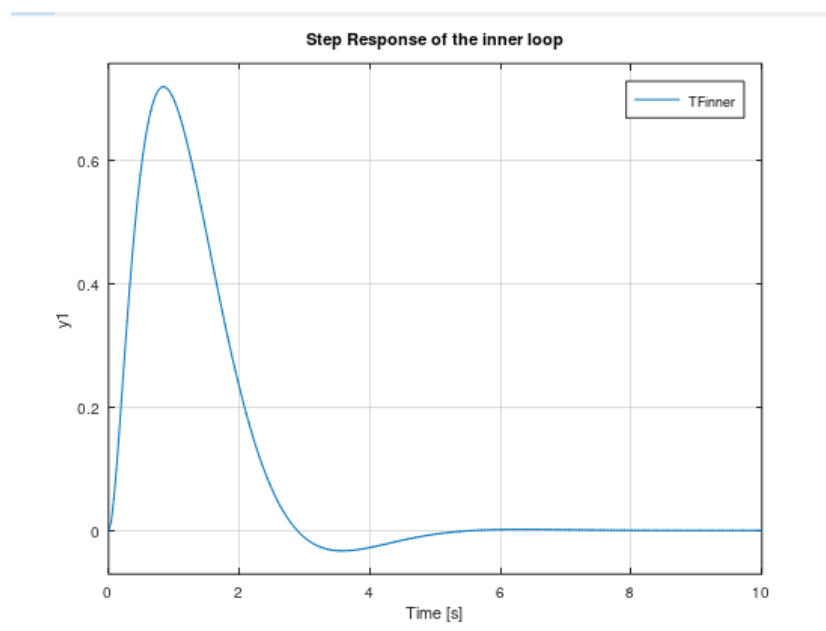
$$\left(U(s) - Y(s) \frac{s + K_1}{s} \right) \frac{20}{s^2 + 10s} = Y(s)$$

$$\frac{Y(s)}{U(s)} = \frac{\frac{20}{s^2 + 10s}}{\left(1 + \frac{20s + 20K_1}{s^3 + 10s^2} \right)}$$

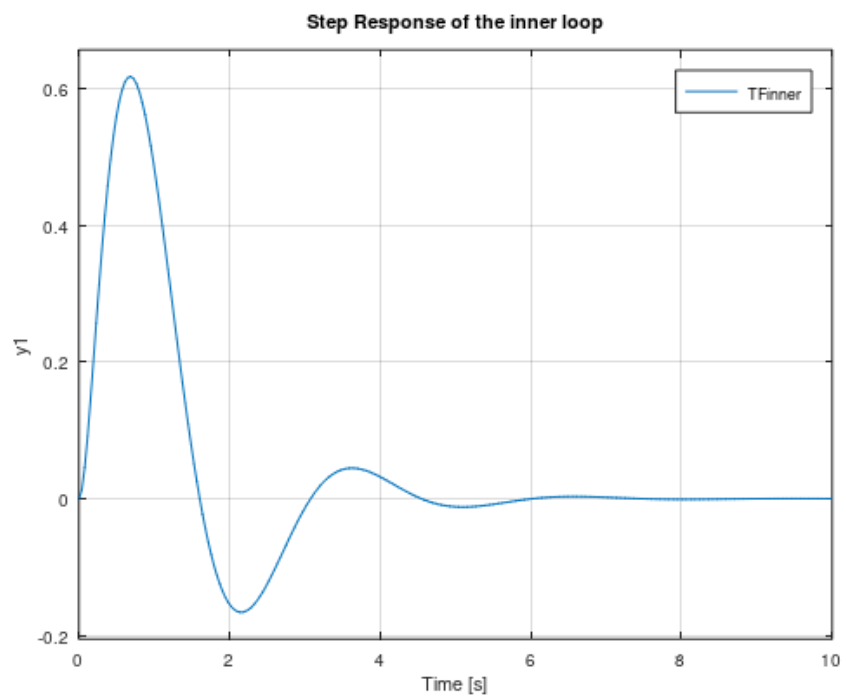
$$TF_{inner}(s) = \frac{Y(s)}{U(s)} = \frac{20s}{s^3 + 10s^2 + 20s + 20K_1}$$

```
>> K1 = [1 2.2 9 10 20]
K1 =
    1.0000    2.2000    9.0000   10.0000   20.0000
```

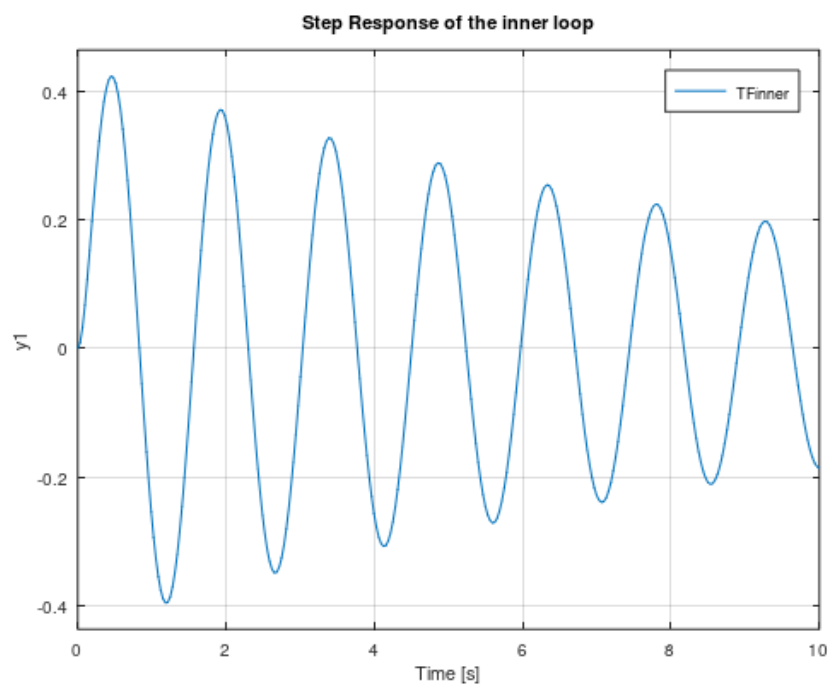
K1= 1



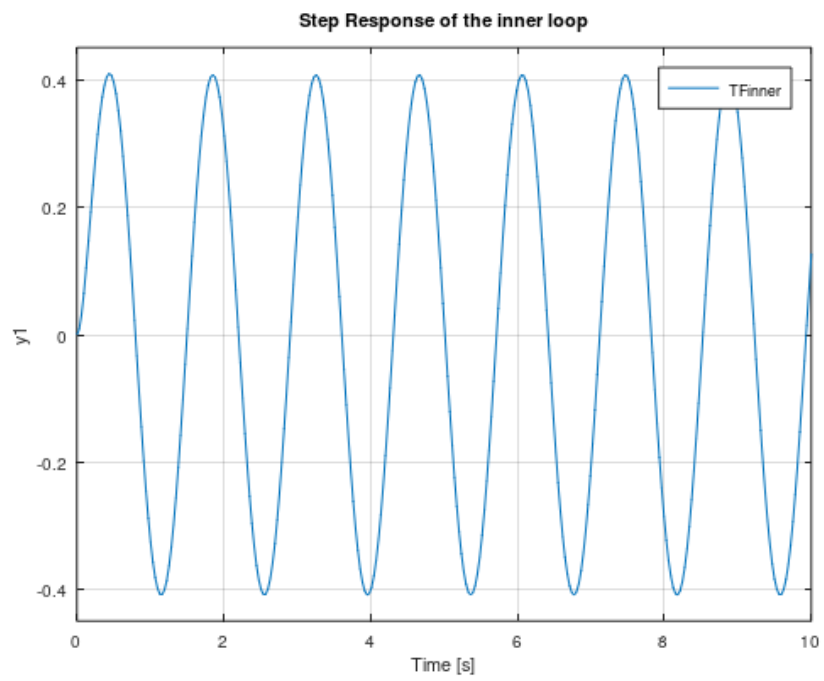
$K_1=2.2$



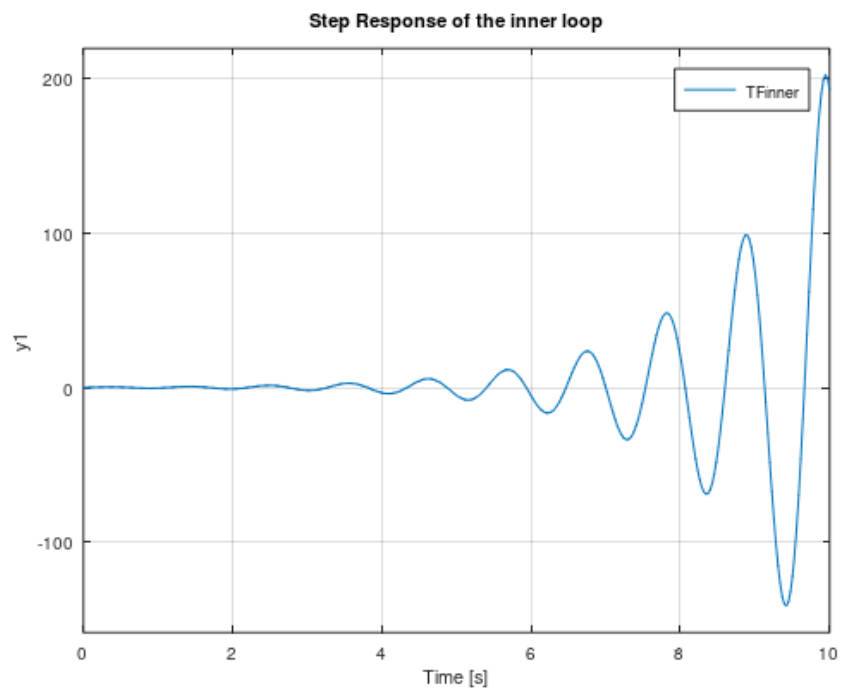
$K_1=9$



$K_1=10$

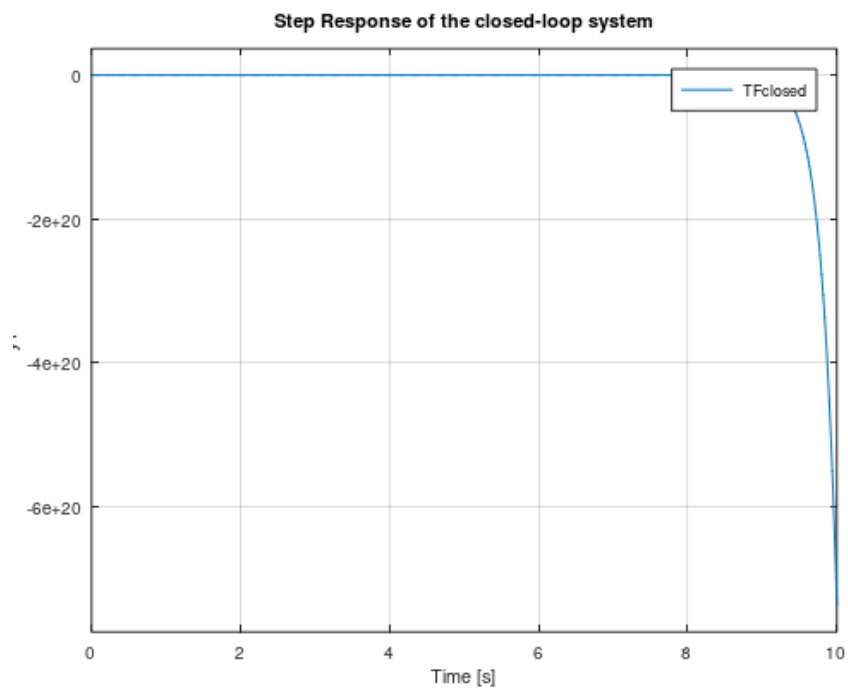


$K_1=20$

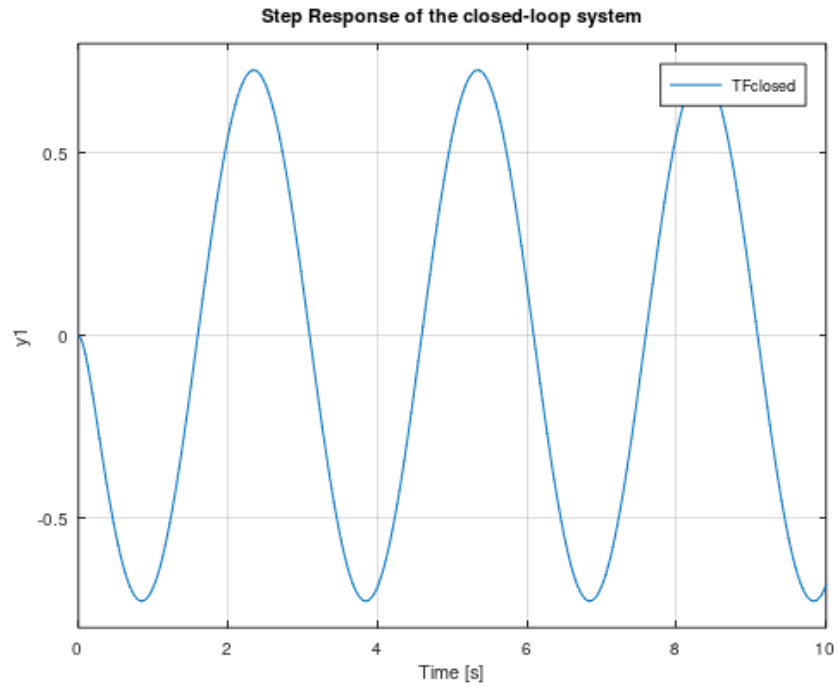


3-

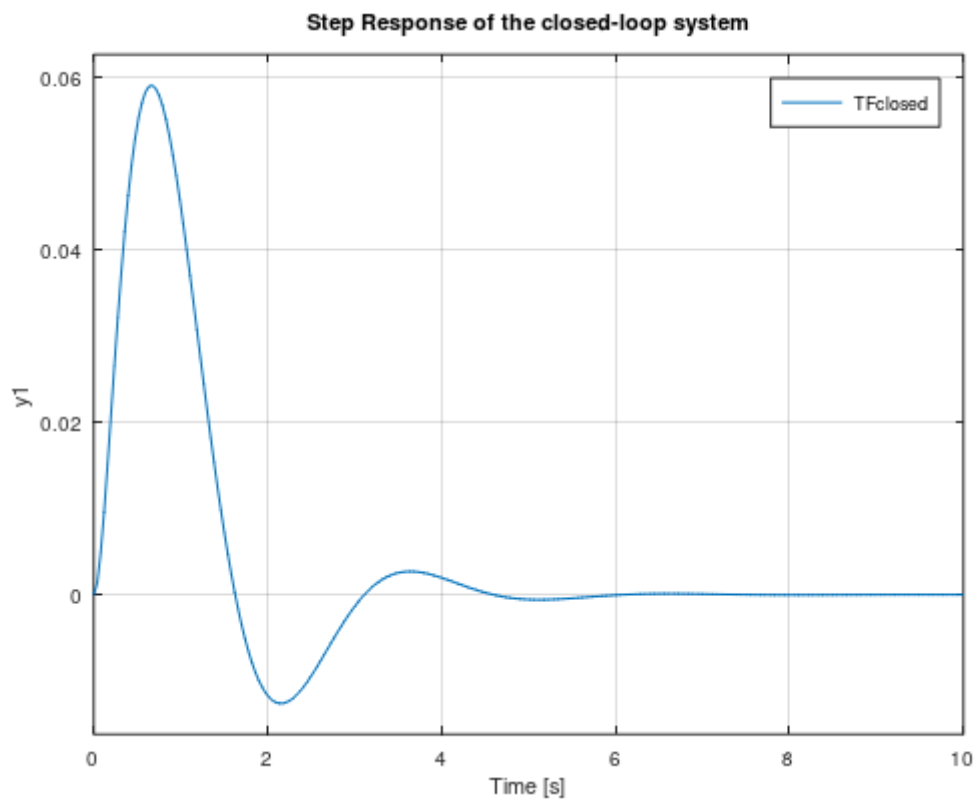
$K_2 = -5$



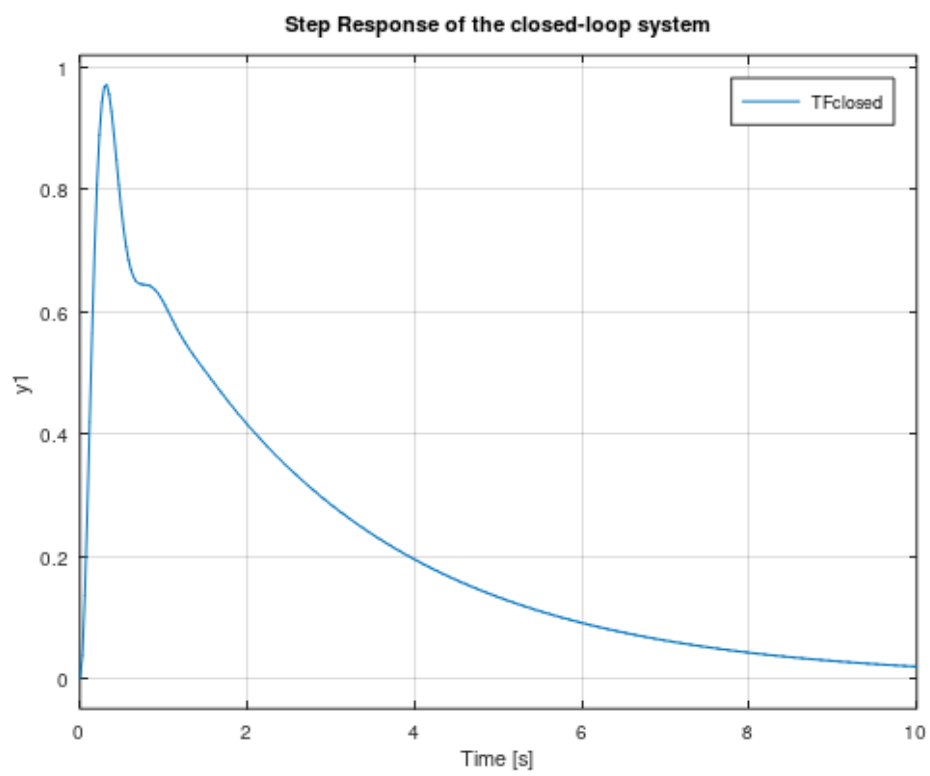
$K_2 = -0.78$



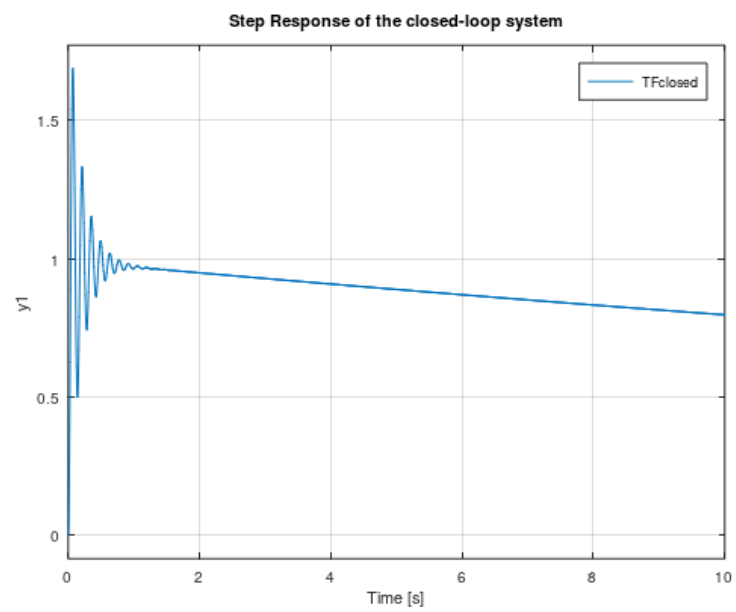
$K_2=0.1$



$K_2=5$



K2=100



B-

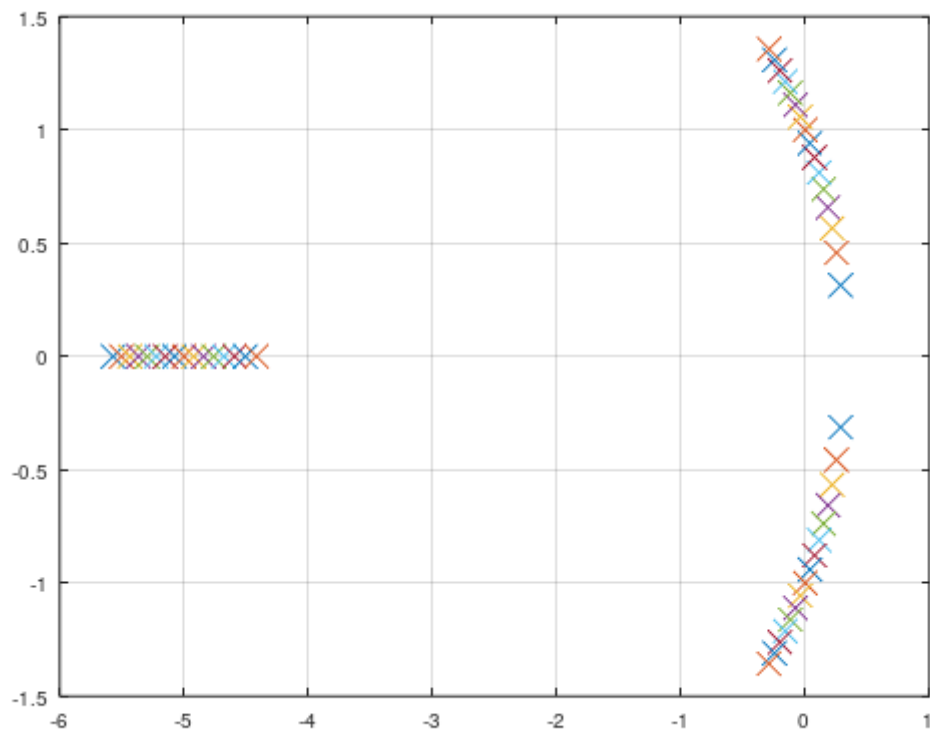
1-

$$TF_{closed}(s) = \frac{Y(s)}{R(s)} = \frac{G(s)}{1 + G(s)} = \frac{1}{s^3 + 5s^2 + (K - 3)s + K + 1}$$

2-

$$K > 4 \quad \text{and} \quad K > -1$$

3- K=4



3. CONCLUSION & DISCUSSION

Discuss your results here. Write down what you learned during the experiment.

For experiment A,

Stable between 0 and 10 values of k_1 .

While the value of k_1 is 2.2, the system is stable when k_2 is above 0.1 value.

For experiment B,

System unstable until k value is 4.

Margin stable at 4.

The system is stable within values of k above 4.

REFERENCES

Write your reference list in this section by using numbering with square brackets - []. You should follow IEEE Citation StyleGuide for referencing format.

- [1] IEEE Citation Style Guide, Available: <http://www.ijssst.info/info/IEEE-Citation-StyleGuide.pdf>, Access Date: October 2021